

Dynamic Threshold Models of Collective Action in Social Networks

Ji Liu

Navid Hassanpour

Sekhar Tatikonda

A. Stephen Morse

Abstract—We intend to understand how decentralized planning through local interactions can result in large-scale change from inactivity to global action cascades characteristic of social revolutions as well as innovation adoption. Building on the dynamic threshold model of threshold learning and collection action proposed in [1], this paper investigates the diffusion process in a number of canonical graph structures with heterogeneous agents. We solve the dynamic equations for star, ring, and infinite path topologies. Of special interest are the situations in which action becomes ubiquitous. When this is not the case, we formulate the *radius of diffusion* as the farthest reach of action from a core of instigators situated in a network of like-minded risk-averse agents. It is shown that full connectivity in a social network can sometimes hinder collective action. Based on the structure of the network and initial thresholds, the model leads the network to one of the possible equilibria of the game on the underlying graph or results in oscillation. Thus the model in this paper predicts non-monotone participation and heterogeneous outcomes, providing a more realistic model of mobilization, innovation diffusion, and opinion reversal.

I. INTRODUCTION

Many social behaviors are known to spread through networks of interacting individuals [2]. With the rapid development of communication technology and fast-growing online social networking services, new means of individual interactions have appeared over the past decade and the nature of the network of interactions has changed greatly [3]. As a result, there has been considerable interest recently in understanding how social networks affect the spread of behaviors [4], [5]. The study of such problems has a long history in sociology [6] and economics [7] and has received much attention lately in the control field [8]–[10].

Similar phenomena also arise in the political context. Historical events such as the Eastern European color revolutions and more recent events such as the post-election unrest in Iran, the anti-government protests in Bangkok, and the 2011 Egyptian revolution are examples of mass collective action resulting from decentralized planning and local mobilization. A common factor among all these political activities is the role of a small number of radicals. It is natural to ask how decentralized planning through local interactions could result in the types of large-scale political unrest characteristic of these events. In [1], a dynamic threshold model of participation in network collective action was proposed in order to analyze

these phenomena. In the present paper, we investigate the solutions to dynamics of this model to find the nature of action cascades in a variety of social structures.

Consider a network consisting of a large number of risk-averse individuals. Each individual is able to communicate only with a subset of other agents in the network which we refer to as his/her “neighbors”. Typically only a small number of individuals in the network try to initiate a demonstration. Each individual in the network has a personal level of risk taking; some are willing to engage in risky behavior by joining a contentious collective action and some are not. We assume that each individual decides to either engage in the demonstration or not based on two factors: his or her personal threshold and the fraction of his or her neighbors who have already joined in. If the fraction is above the threshold, the individual in question joins the demonstration; otherwise he or she does not. This well-known threshold model of collective action was first proposed in [6], [11] and later expanded in [12]–[15].

A variety of assumptions have been made about thresholds. For example, in [16] the thresholds of all the individuals in the network are assumed to be fixed and equal. However, the homogeneity assumption is limiting. For instance, it is clear that the relation between radicals and non-radicals demands a model with at least two classes of thresholds. In addition, it is plausible to think that when participating becomes prevalent, one’s threshold could decrease; or as inaction becomes the norm, thresholds could also increase. We capture these considerations in our stylization.

Dynamic models of thresholds as an explanation for cascades of collective action are examined in [12], [17]. Later similar dynamic threshold models were studied in [14], [15], [18]. In [13] threshold updates are assumed to be Bayesian. Following [1], [19], we adopt a local averaging model which exploits the idea of flocking [20]. This model is a special case of the DeGroot model widely used in social learning [21].

II. THE MODEL

We consider a social network consisting of $n > 1$ individuals labeled 1 to n . Each individual i is able to communicate only with certain other individuals in the network called individual i ’s *neighbors*. For simplicity we take each individual to be a neighbor of himself or herself. Neighbor relations are described by an undirected graph $\mathbb{G}$, called a *neighbor graph*, in which vertices correspond to individuals and edges indicate neighbor relations. Thus $\mathbb{G}$ has self-loops at all vertices and the neighbors of individual i have the same labels as the vertices in $\mathbb{G}$ which are adjacent to vertex i . In this paper we consider the case when neighbor

The research of J. Liu and A. S. Morse is supported by the US Air Force Office of Scientific Research and the National Science Foundation.

N. Hassanpour is with the Department of Political Science, Yale University, USA (navid.hassanpour@yale.edu). J. Liu, S. Tatikonda, and A. S. Morse are with the Department of Electrical Engineering, Yale University, USA ({ji.liu, sekhar.tatikonda, as.morse}@yale.edu).

relations are fixed for all time. Thus $\mathbb{G}$ does not depend on time.

Each individual i has control over a real-valued scalar variable θ_i , called a *threshold*, which takes values in the real closed interval $[0, 1]$, and a binary-valued variable a_i , called an *action variable*, which individual i is able to update from time to time. The value of $a_i(t)$ can be either one or zero. If $a_i(t) = 1$, individual i is said to be *active* at time t whereas if $a_i(t) = 0$, individual i is *inactive* at time t . For individual i to be *active* means that individual i is engaged in a demonstration [1].

Initially the threshold values of all n individuals are taken to be given numbers between 0 and 1. It is assumed that the initial value of action variable $a_i(0)$ is 1 if $\theta_i(0) = 0$, or $a_i(0) = 0$ if $\theta_i(0) \neq 0$. A individual i is said to be a *radical* if its initial threshold value $\theta_i(0) = 0$. Otherwise, individual i is called *ordinary*. Typically only a small number of individuals in the group are initially radical. We are interested in understanding how the model predicts the spread of activism throughout the network.

In accordance with the dynamic model first proposed in [1], all n individuals update their thresholds and action variables in the following manner. At each time $t \in \{0, 1, \dots\}$ each individual i first updates its threshold using the rule

$$\theta_i(t+1) = \frac{1}{n_i} \sum_{j \in \mathcal{N}_i} \theta_j(t) \quad (1)$$

where n_i is the number of neighbors of agent i and $\mathcal{N}_i$ is the set of labels of the neighbors of agent i . Thus individual i updates its threshold using a local rule based on the average of the thresholds of his or her neighbors. Then individual i computes the proportion of his or her neighbors who are active at time t :

$$p_i(t) = \frac{m_i(t)}{n_i} \quad (2)$$

where $m_i(t)$ is the number of neighbors of individual i who are active at time t . Finally individual i updates its action variable using the rule

$$a_i(t+1) = \begin{cases} 1 & \text{if } p_i(t) \geq \theta_i(t+1) \\ 0 & \text{if } p_i(t) < \theta_i(t+1) \end{cases} \quad (3)$$

The preceding set of update rules defined by (1)–(3) can be written in state form. Toward this end, define

$$F = D^{-1}A$$

where A is the adjacency matrix of graph $\mathbb{G}$ and D is the diagonal matrix whose i th diagonal entry is the degree of vertex i within the graph. Then

$$\theta(t+1) = F\theta(t) \quad (4)$$

$$a(t+1) = \phi(Fa(t) - F\theta(t)) \quad (5)$$

where $\theta = [\theta_1 \ \theta_2 \ \dots \ \theta_n]'$ and $a = [a_1 \ a_2 \ \dots \ a_n]'$ are respectively the vector of thresholds and action variables and $\phi: [-1, 1] \rightarrow \{0, 1\}$ is the function defined so that $\phi(x) = 0$ if $x < 0$ and $\phi(x) = 1$ if $x \geq 0$. It is worth noting that F is a stochastic matrix [20].

III. POSSIBLE EQUILIBRIA

The aim of this section is to discuss the equilibria of the system (4) and (5). First, it is well known that in the case when $\mathbb{G}$ is connected,

$$\lim_{t \rightarrow \infty} \theta(t) = \theta_e \mathbf{1}$$

where $\mathbf{1} = [1 \ 1 \ \dots \ 1]'$ and θ_e is a number depending only on $\mathbb{G}$ and $\theta(0)$ [20]. It can be verified that

$$\theta_e = \frac{\sum_{k=1}^n n_k \theta_k(0)}{\sum_{k=1}^n n_k} \quad (6)$$

Thus it is clear that $0 \leq \theta_e \leq 1$.

Now we consider possible limit points of a . Suppose that a has a limit a_e as $t \rightarrow \infty$. Then

$$a_e = \phi(Fa_e - \theta_e \mathbf{1}) \quad (7)$$

It is easy to see that for any graph $\mathbb{G}$, $a_e = \mathbf{1}$ and $a_e = \mathbf{0}$ are two possible equilibria of (7) where $\mathbf{0} = [0 \ 0 \ \dots \ 0]'$.

If a_e is neither $\mathbf{1}$ nor $\mathbf{0}$, more can be said. We say that an individual i is *isolated* if the limiting value of his or her action variable is different than that of all of his or her neighbors. More precisely, an individual i is an *isolated active individual* if the limit of his or her action variable is one and the limit of his or her neighbors' action variables all equal zero; an individual is an *isolated inactive individual* if the limit of his or her limit action variable is zero and the limit of his or her neighbors' action variables all equal one. A vertex i in $\mathbb{G}$ is called an *isolated active vertex* if individual i is isolated active; vertex i is an *isolated inactive vertex* if individual i is isolated inactive.

Lemma 1: There does not exist a neighbor graph $\mathbb{G}$ with both isolated active and isolated inactive vertices. Moreover, if $\mathbb{G}$ is a regular graph, then $\mathbb{G}$ cannot have any isolated vertices.

The proof of Lemma 1 is omitted due to space limitation and will be given in a full length version of this paper.

IV. ASYMPTOTIC BEHAVIORS

In the sequel we will investigate the dynamic behavior of the system (4) and (5) for some special classes of neighbor graphs in the case when there is a single radical in the network.

A. Complete Graphs

Consider a complete graph with $n > 1$ vertices. Suppose that at initial time $t = 0$, $\theta_1(0) = 0$ and $\theta_i(0) = \tau$, $0 < \tau < 1$, for all $i \in \{2, 3, \dots, n\}$. Thus individual 1 is the unique radical one. In this case the initial value of action variable of individual 1 is one and the initial values of action variables of the remaining individuals are all zeros.

First note that $p_i(0) = \frac{1}{n}$ for each $i \in \{1, 2, \dots, n\}$. In addition, it is easy to verify that all n thresholds reach a consensus in one iterative step. Thus for all $t \geq 1$ and $i \in \{1, 2, \dots, n\}$,

$$\theta_i(t) = \left(\frac{n-1}{n} \right) \tau$$

Then it is clear that if $\tau \leq \frac{1}{n-1}$, all the action variables equal one at time $t = 1$, which is a steady state of the system; if $\tau > \frac{1}{n-1}$, all the action variables equal zero at time $t = 1$, which is also a steady state of the system. We are led to the following result.

Proposition 1: Suppose that $\mathbb{G}$ is a complete graph with $n > 1$ vertices and that all n individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of a single radical whose initial threshold is zero. Then all n individuals will be active in one iterative step if $\tau \leq \frac{1}{n-1}$ and all n individuals will be inactive in one iterative step if $\tau > \frac{1}{n-1}$.

B. Star Graphs

Consider a star graph with $n+1$ vertices, $n > 1$. The centre is labeled 0 and the leaves are labeled 1 to n .¹ Suppose that the centre is the unique radical one and that the initial values of the thresholds of all leaves are the same. Then it is easy to see that the values of thresholds, as well as the values of action variables, of all leaves are equal all the time. We use θ_c, a_c to respectively denote the threshold and action variable of the centre 0 and θ_l, a_l to respectively denote the threshold and action variable of the leaves 1 to n . At initial time $t = 0$, $\theta_c(0) = 0$ since centre 0 is radical; we assume that $\theta_l(0) = \tau$, $0 < \tau < 1$. Then the initial value of the centre's action variable is one and the initial values of action variables of the leaves all equal zero.

Lemma 2: Suppose that $\mathbb{G}$ is a star graph with $n+1$ vertices and that the vector of thresholds $\theta(t)$ evolves according to (4). Let the initial values of all the thresholds equal τ with the exception of the central one whose initial value is zero. Then

$$\theta_c(t) = \left(\frac{2n(2n+2)^t - 2n(1-n)^t}{(3n+1)(2n+2)^t} \right) \tau \quad (8)$$

$$\theta_l(t) = \left(\frac{4n(2n+2)^{t-1} + (1-n)^t}{2(3n+1)(2n+2)^{t-1}} \right) \tau \quad (9)$$

Lemma 2 can be proved by induction on t . The simple proof is omitted.

By (8) and (9), it is clear that both θ_c and θ_l will converge to

$$\theta_e = \frac{2n\tau}{3n+1}$$

as $t \rightarrow \infty$, which is consistent with (6). Now we turn to the dynamics of the vector of action variables a .

By symmetry, the values of all leaves' action variables are equal for all $t \geq 0$. Thus there are only four possible states of a :

- 1) $a_c = 0$ and $a_l = 1$
- 2) $a_c = 1$ and $a_l = 0$
- 3) $a_c = 0$ and $a_l = 0$
- 4) $a_c = 1$ and $a_l = 1$

¹In a star graph with $n+1$ vertices, $n > 1$, the unique vertex of degree greater than one is called the *centre* of the star and the remaining n vertices of degree one are called the *leaves* of the star.

It is easy to see that the first one is the initial value of a and the last two are both steady states. More can be said.

Lemma 3: Suppose that $\mathbb{G}$ is a star graph with $n+1$ vertices and that all $n+1$ individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of the central radical whose initial threshold is zero. Then when time $t > 0$ is odd, $a_c(t) = 0$ and $a_l(t) = 1$ if and only if

$$\begin{aligned} \tau &\leq \frac{(3n+1)(2n+2)^{t-1}}{4n(2n+2)^{t-1} + (1-n)^t} \\ \tau &> \frac{(3n+1)(2n+2)^{t-2}}{4n(2n+2)^{t-2} + (1-n)^{t-1}} \end{aligned}$$

When time $t > 0$ is even, $a_c(t) = 1$ and $a_l(t) = 0$ if and only if

$$\begin{aligned} \tau &\leq \frac{(3n+1)(2n+2)^{t-2}}{4n(2n+2)^{t-2} + (1-n)^{t-1}} \\ \tau &> \frac{(3n+1)(2n+2)^{t-1}}{4n(2n+2)^{t-1} + (1-n)^t} \end{aligned}$$

Proof of Lemma 3: The lemma will be proved by induction on t . First it is easy to verify that the lemma is true when $t = 1$ and $t = 2$.

Now suppose that the lemma holds for $t = m$ where $m \geq 1$ is a positive integer. Let $t = m+1$. Time $m+1$ may be either odd or even. We consider these two cases separately.

Suppose that m is even in which case $m = 2k$, $k \in \{1, 2, \dots\}$, and that $a_c(m) = 1$, $a_l(m) = 0$. Then

$$p_c(m) = \frac{1}{n+1} \quad \text{and} \quad p_l(m) = \frac{1}{2}$$

By the inductive hypothesis,

$$\tau > \frac{(3n+1)(2n+2)^{2k-1}}{4n(2n+2)^{2k-1} + (1-n)^{2k}}$$

By Lemma 2,

$$\theta_c(m+1) = \left(\frac{2n(2n+2)^{2k+1} - 2n(1-n)^{2k+1}}{(3n+1)(2n+2)^{2k+1}} \right) \tau$$

$$\theta_l(m+1) = \left(\frac{4n(2n+2)^{2k} + (1-n)^{2k+1}}{2(3n+1)(2n+2)^{2k}} \right) \tau$$

It can be verified that $p_c(m) < \theta_c(m+1)$, which implies that $a_c(m+1) = 0$. Thus if $p_l(m) < \theta_l(m+1)$, then $a_l(m+1) = a_c(m+1) = 0$, which is a steady state. On the other hand, if $p_l(m) \geq \theta_l(m+1)$, or equivalently,

$$\tau \leq \frac{(3n+1)(2n+2)^{2k}}{4n(2n+2)^{2k} + (1-n)^{2k+1}}$$

then $a_c(m+1) = 0$, $a_l(m+1) = 1$.

Next suppose that m is odd in which case $m = 2k+1$, $k \in \{1, 2, \dots\}$, and that $a_c(m) = 0$, $a_l(m) = 1$. Then

$$p_c(m) = \frac{n}{n+1} \quad \text{and} \quad p_l(m) = \frac{1}{2}$$

By the inductive hypothesis,

$$\tau \leq \frac{(3n+1)(2n+2)^{2k}}{4n(2n+2)^{2k} + (1-n)^{2k+1}}$$

By Lemma 2,

$$\theta_c(m+1) = \left(\frac{2n(2n+2)^{2k+2} - 2n(1-n)^{2k+2}}{(3n+1)(2n+2)^{2k+2}} \right) \tau$$

$$\theta_l(m+1) = \left(\frac{4n(2n+2)^{2k+1} + (1-n)^{2k+2}}{2(3n+1)(2n+2)^{2k+1}} \right) \tau$$

It can be verified that $p_c(m) \geq \theta_c(m+1)$, which implies that $a_c(m+1) = 1$. Thus if $p_l(m) \geq \theta_l(m+1)$, then $a_l(m+1) = a_c(m+1) = 1$, which is a steady state. On the other hand, if $p_l(m) < \theta_l(m+1)$, or equivalently,

$$\tau > \frac{(3n+1)(2n+2)^{2k+1}}{4n(2n+2)^{2k+1} + (1-n)^{2k+2}}$$

then $a_c(m+1) = 1$, $a_l(m+1) = 0$. By induction, the lemma is established and the proof is complete. ■

By Lemma 3 it is clear that the vector of action variables a will converge to a steady state, either **1** or **0**, unless τ is located in a time-varying interval, which is

$$\begin{aligned} \tau &> \frac{(3n+1)(2n+2)^{2k-1}}{4n(2n+2)^{2k-1} + (1-n)^{2k}} \\ \tau &\leq \frac{(3n+1)(2n+2)^{2k-2}}{4n(2n+2)^{2k-2} + (1-n)^{2k-1}} \end{aligned}$$

It is easy to verify that for $k \in \{1, 2, \dots\}$, the lower endpoint of the interval is an increasing sequence and the upper endpoint of the interval is a decreasing sequence. Thus the length of the interval is shrinking. Note that for each finite $k \in \{1, 2, \dots\}$

$$\begin{aligned} \frac{3n+1}{4n} &> \frac{(3n+1)(2n+2)^{2k-1}}{4n(2n+2)^{2k-1} + (1-n)^{2k}} \\ \frac{3n+1}{4n} &\leq \frac{(3n+1)(2n+2)^{2k-2}}{4n(2n+2)^{2k-2} + (1-n)^{2k-1}} \end{aligned}$$

In addition, both the lower and upper endpoints of the above interval converge to $\frac{3n+1}{4n}$ as $k \rightarrow \infty$. We are led to the following result.

Proposition 2: Suppose that $\mathbb{G}$ is a star graph with $n+1$ vertices and that all $n+1$ individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of the central radical whose initial threshold is zero. Then all the individuals will be active in finite time if $\tau < \frac{3n+1}{4n}$ and all the individuals will be inactive in finite time if $\tau > \frac{3n+1}{4n}$. If $\tau = \frac{3n+1}{4n}$, then the vector of action variables a will oscillate between two states.

C. Ring Graphs

Consider a ring graph with $n+1$ vertices, $n > 1$. The vertices are labeled 0 to n clockwise. Suppose that at initial time $t = 0$, $\theta_0(0) = 0$ and $\theta_i(0) = \tau$, $0 < \tau < 1$, for all $i \in \{1, 2, \dots, n\}$. Thus individual 0 is the unique radical one.

In this case the initial value of action variable of vertex 0 is one and the initial values of action variables of the remaining vertices are all zeros.

Since the network is symmetric with respect to vertex 0, the dynamics of vertex i is the same as vertex $(n+1-i)$, $i \in \{1, 2, \dots, m\}$, where

$$m = \left\lfloor \frac{n}{2} \right\rfloor$$

Here $\lfloor \cdot \rfloor$ denotes the floor function defined so that $\lfloor x \rfloor$ equals the largest integer not greater than x . We list some values of thresholds in the following table:

	0	1	2	3	4
$t = 0$	0	τ	τ	τ	τ
$t = 1$	$\frac{2}{3}\tau$	$\frac{2}{3}\tau$	τ	τ	τ
$t = 2$	$\frac{6}{9}\tau$	$\frac{7}{9}\tau$	$\frac{8}{9}\tau$	τ	τ
$t = 3$	$\frac{20}{27}\tau$	$\frac{21}{27}\tau$	$\frac{24}{27}\tau$	$\frac{26}{27}\tau$	τ
$t = 4$	$\frac{62}{81}\tau$	$\frac{65}{81}\tau$	$\frac{71}{81}\tau$	$\frac{77}{81}\tau$	$\frac{80}{81}\tau$

From the table, it is easy to see that if $\tau > \frac{1}{2}$, then $a_i(1) = 0$ for all $i \in \{0, 1, 2, \dots, n\}$, which is a steady state. More can be said:

- 1) For all $i \in \{0, 1, 2, \dots, n\}$,

$$\lim_{t \rightarrow \infty} \theta_i(t) = \frac{n\tau}{n+1}$$

- 2) For all $t \geq 0$, $\theta_0(t) \leq \theta_1(t) \leq \theta_2(t) \leq \dots \leq \theta_m(t)$.
- 3) $\theta_0(t)$ is a nondecreasing sequence of time t .
- 4) For any $t \geq 1$, at most one vertex in $\{1, 2, \dots, m\}$ which changes the value of its action variable from 0 to 1.
- 5) For each $k \in \{1, 2, \dots, m\}$, vertex k changes the value of its action variable from 0 to 1 at time $t = k$ if and only if

$$\tau \leq \frac{3^{k-1}}{3^k - 1}$$

The simple proofs of the preceding facts are omitted.

In the sequel we assume that $\tau \leq \frac{1}{2}$. In this case, vertex 1 changes its action variable from 0 to 1 and all the remaining vertices keep inactive at time $t = 1$. Then $p_1 = \frac{2}{3}$. But $\theta_1 = \frac{2}{3}\tau$ at time $t = 1$ and will never exceed τ . Thus $p_1(t) > \theta_1(t)$ for all $t \geq 1$, which implies that vertex 1 will never change its action variable from 1 to 0. By using the same arguments, one reaches the conclusion that if a vertex changes its action variable from 0 to 1, then it will never change it back to 0 from 1.

Suppose at some time $t \geq 1$, vertex i is active and vertex $i+1$ is inactive, $i \in \{1, 2, \dots, m\}$. From the preceding it is clear that all vertices in $\{0, 1, 2, \dots, i-1\}$ have been active and will never change back to inaction. Thus the limit point

of a is either 1 or in the form of

$$\begin{aligned} a_0 &= 1, \\ a_i &= a_{n+1-i} = 1, \quad i \in \{1, 2, \dots, r\} \\ a_j &= 0, \quad j \in \{r+1, r+2, \dots, n-r\} \end{aligned}$$

where $r \leq m$. We call r the *radius of diffusion*.

Suppose n is even in which case $n = 2m$. By fact 5, it is easy to see that $a(t) = 1$ for all $t \geq m$ if

$$\tau \leq \frac{3^{m-1}}{3^m - 1} \quad (10)$$

which implies that $r = m$. Next suppose that n is odd in which case $n = 2m + 1$. By fact 5 it is clear that $a_0 = 1$, $a_i = a_{n+1-i} = 1$, $i \in \{1, 2, \dots, m\}$, and $a_{m+1} = 0$ at time $t = m$ if (10) holds. At time $t = m + 1$, it can be verified that $a_{m+1} = 1$ if and only if

$$\tau \leq \frac{2 \cdot 3^m}{3^{m+1} - 2}$$

which must hold if (10) holds. Thus $a(t) = 1$ for all $t \geq m + 1$.

We summarize as follows.

Proposition 3: Suppose that $\mathbb{G}$ is a ring graph with $n + 1$ vertices and that all $n + 1$ individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of a single radical whose initial threshold is zero. Then all the individuals will be inactive in one iterative step if $\tau > \frac{1}{2}$ and all the individuals will be active in finite time if

$$\tau \leq \frac{3^{m-1}}{3^m - 1}$$

where $m = \lfloor \frac{n}{2} \rfloor$.

In the case when

$$\frac{3^{m-1}}{3^m - 1} < \tau \leq \frac{1}{2}$$

The explicit relation between τ and the radius of diffusion r has so far eluded us. We plot this relation in Figure 1 by simulation.

D. Path Graphs

First we consider a path graph with odd number of vertices and one single radical which is located in the middle of the path.

Proposition 4: Suppose that $\mathbb{G}$ is a path graph with $2n + 1$ vertices and that all $2n + 1$ individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of the central radical whose initial threshold is zero. Then all the individuals will be inactive in one iterative step if $\tau > \frac{1}{2}$ and all the individuals will be active in finite time if

$$\tau \leq \frac{3^{n-2}}{3^{n-1} - 1}$$

This proposition can be proved using the same arguments in the preceding section.

In the case when the number of vertices is infinite, more can be said.

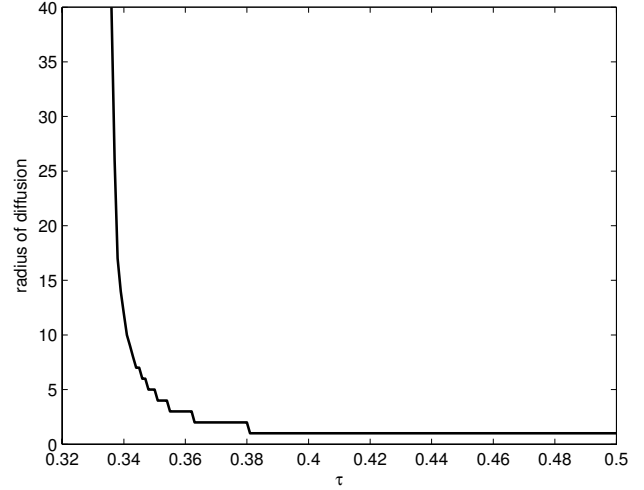


Fig. 1. Relation between τ and the radius of diffusion r ($n = 100$)

Consider an infinite path graph with one single radical. The radical vertex is labeled 0 and remaining vertices are labeled as Figure 2. We use the symbol θ_i to denote the threshold of vertex i . At initial time $t = 0$, we assume that $\theta_0 = 0$ and $\theta_i = \tau$, $i \neq 0$, $0 < \tau < 1$. In this case the initial value of action variable of vertex 0 is one and the initial values of action variables of remaining vertices all equal zero.

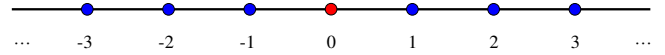


Fig. 2. Infinite path with one radical vertex (in red)

Lemma 4: Suppose that $\mathbb{G}$ is an infinite path graph labeled as Figure 2 and that the vector of thresholds $\theta(t)$ evolves according to (4). Let the initial values of all the thresholds equal τ with the exception of vertex 0 whose initial value is zero. Then for all $i \in \{0, \pm 1, \pm 2, \dots\}$,

$$\lim_{t \rightarrow \infty} \theta_i(t) = \tau$$

The proof of Lemma 4 is omitted due to space limitation and will be given in a full length version of this paper.

It is easy to see that if $\tau > \frac{1}{2}$, then all vertices, including the radical one, are inactive at time $t = 1$, which is a steady state.

In the sequel we assume that $\tau \leq \frac{1}{2}$. In this case, vertex 1 changes its action variable from 0 to 1 and all the remaining vertices keep inactive at time $t = 1$. Then $p_1 = \frac{2}{3}$. But $\theta_1 = \frac{2}{3}\tau$ at time $t = 1$ and will never exceed τ . Thus $p_1(t) > \theta_1(t)$ for all $t \geq 1$, which implies that vertex 1 will never change its action variable from 1 to 0. By using the same arguments, one reaches the conclusion that if a vertex changes its action variable from 0 to 1, then it will never change it back to 0 from 1. Moreover it can be verified that for each $k \in$

$\{1, 2, \dots\}$, vertex k changes the value of its action variable from 0 to 1 at time $t = k$ if and only if

$$\tau \leq \frac{3^{k-1}}{3^k - 1}$$

Suppose at some time $t \geq 1$, vertex i is active and vertex $i + 1$ is inactive, $i \in \{1, 2, \dots\}$. From the preceding it is clear that all vertices in $\{0, 1, 2, \dots, i - 1\}$ have been active and will never change back to inaction. Then $p_{i+1}(t) = \frac{1}{3}$. Recall that $\theta_{i+1}(t)$ converges to τ as $t \rightarrow \infty$. Thus if $\tau \leq \frac{1}{3}$, then vertex $i + 1$ will be active at time $t + 1$. By repeating this argument for all $i \in \{1, 2, \dots\}$, all vertices will be active as $t \rightarrow \infty$ if $\tau \leq \frac{1}{3}$.

Now we will study how the system behave in the case when $\frac{1}{3} < \tau \leq \frac{1}{2}$. It is worth noting that

$$\left\{ \frac{3^{k-1}}{3^k - 1}, k \in \{1, 2, \dots\} \right\}$$

is a decreasing sequence. In particular,

$$\frac{3^0}{3^1 - 1} = \frac{1}{2} \quad \text{and} \quad \lim_{k \rightarrow \infty} \frac{3^{k-1}}{3^k - 1} = \frac{1}{3}$$

For any τ such that $\frac{1}{3} < \tau \leq \frac{1}{2}$, there must exist an integer m such that

$$\frac{3^m}{3^{m+1} - 1} < \tau \leq \frac{3^{m-1}}{3^m - 1}$$

Thus vertex m is active at time $t = m$ and vertex $m + 1$ is inactive at time $t = m + 1$. Let μ be a real number such that $\frac{1}{3} < \mu < \tau$. Since $\lim_{t \rightarrow \infty} \theta_0(t) = \tau$, there must exist an integer q such that $\theta_0(t) > \mu > \frac{1}{3}$ for all $t \geq q$. Since $\theta_0(t) \leq \theta_1(t) \leq \theta_2(t) \leq \dots$ for all $t \geq 0$, $\theta_i(t) > \mu > \frac{1}{3}$ for all $i \in \{0, 1, 2, \dots\}$ and $t \geq q$. Note that at each time clock, at most one vertex in $\{1, 2, \dots\}$ changes the value of its action variable from 0 to 1. Thus up to time $t = q$, at most $2q + 1$ vertices have changed their action variables from 0 to 1 and they will never change back to 0. But from then on, no more vertices will change their action variables from 0 to 1 since $\theta_i > \frac{1}{3}$ for all i .

We summarize:

Proposition 5: Suppose that $\mathbb{G}$ is an infinite path graph labeled as Figure 2 and that all individuals follow the dynamic model (4) and (5). Let the initial thresholds of all the individuals equal τ , $0 < \tau < 1$, with the exception of individual 0 whose initial threshold is zero. Then all the individuals will be inactive in one iterative step if $\tau > \frac{1}{2}$, all the individuals will be active as time $t \rightarrow \infty$ if $\tau \leq \frac{1}{3}$, and finite number of individuals will be active as time $t \rightarrow \infty$ if $\frac{1}{3} < \tau \leq \frac{1}{2}$.

V. CONCLUDING REMARKS

A dynamic threshold model is studied in this paper to understand how the structure of a social networks affect the spread of collective action through local interactions. It is shown that network structure and individuals' initial levels of risk preference can critically affect the extent to which political mobilization spreads across networks of interacting individuals.

Sometimes the information on neighbors' thresholds is hard to exchange in the course of fast-paced interactions. Hence individuals in the network have to infer the real values of their neighbors' thresholds from the previous action history of their neighbors. One variation of this case has been studied in [1]. An analysis of such models is a subject for future research.

VI. ACKNOWLEDGMENT

The authors wish to thank Bruce Francis, Mohamed-Ali Belabbas, Brian D. O. Anderson, and Yu-Wen Hsu for useful discussions which have contributed to this work.

REFERENCES

- [1] N. Hassanpour. Media disruption exacerbates revolutionary unrest: Evidence from Mubarak's natural experiment. In *Proceedings of the American Political Science Association Annual Meeting*, 2011. Available at SSRN: <http://ssrn.com/abstract=1903351>.
- [2] E. M. Rogers. *Diffusion of Innovations*. The Free Press, New York, NY, 1995.
- [3] N. B. Ellison, C. Steinfield, and C. Lampe. The benefits of Facebook "friends": social capital and college students' use of online social network sites. *Journal of Computer-Mediated Communication*, 12(4):1143–1168, 2007.
- [4] D. Centola. The spread of behavior in an online social network experiment. *Science*, 329(5996):1194–1197, 2010.
- [5] A. Montanari and A. Saberi. The spread of innovations in social networks. *Proceedings of the National Academy of Science*, 107(47):20196–20201, 2010.
- [6] M. S. Granovetter. Threshold models of collective behavior. *The American Journal of Sociology*, 83(6):1420–1443, 1978.
- [7] P. A. David. Clio and the economics of QWERTY. *The American Economic Review*, 75(2):332–337, 1985.
- [8] D. Acemoglu, A. Ozdaglar, and E. Yildiz. Diffusion of innovations in social networks. In *Proceedings of the 50th IEEE Conference on Decision and Control*, pages 2329–2334, 2011.
- [9] P. Molavi and A. Jadbabaie. Aggregate observational distinguishability is necessary and sufficient for social learning. In *Proceedings of the 50th IEEE Conference on Decision and Control*, pages 2335–2340, 2011.
- [10] D. Acemoglu, G. Como, F. Fagnani, and A. Ozdaglar. Opinion fluctuations and persistent disagreement in social networks. In *Proceedings of the 50th IEEE Conference on Decision and Control*, pages 2347–2352, 2011.
- [11] T. C. Schelling. *Micromotives and Macrobehavior*. W. W. Norton & Company, New York, NY, 1978.
- [12] T. Kuran. Sparks and prairie fires: A theory of unanticipated political revolution. *Public Choice*, 61(0):41–74, 1989.
- [13] S. Lohmann. The dynamics of informational cascades: The Monday demonstrations in Leipzig, East Germany, 1989-91. *World Politics*, 47(1):42–101, 1994.
- [14] R. V. Gould. Collective action and network structure. *American Sociological Review*, 58(2):182–196, 1993.
- [15] D. A. Siegel. Social networks and collective action. *American Journal of Political Science*, 53(1):122–138, 2009.
- [16] S. Morris. Contagion. *Review of Economics Studies*, 67(1):57–78, 2000.
- [17] T. Kuran. Chameleon voters and public choice. *Public Choice*, 53(1):53–78, 1987.
- [18] M. Chwe. Structure and strategy in collective action. *American Journal of Sociology*, 105(1):128–156, 1999.
- [19] N. Hassanpour. Dynamic models of mobilization in political networks. In *Proceedings of the 2010 Political Networks Conference*, 2010. Available at SSRN: <http://ssrn.com/abstract=1669823>.
- [20] A. Jadbabaie, J. Lin, and A. S. Morse. Coordination of groups of mobile autonomous agents using nearest neighbor rules. *IEEE Transactions on Automatic Control*, 48(6):988–1001, 2003.
- [21] M. O. Jackson. *Social and Economic Networks*. Princeton University Press, Princeton, NJ, 2008.